

100X IMPROVEMENT IN BIOELECTRONIC SIGNAL ACQUISITION

W/L = 1000 resonant ladder

AC sweep, per-channel zooms, and a noise-driven time-domain confirmation

Configuration

device W/L = 1000 (W = 1000 um, L = 1 um)
model Kp = 0.717 uA/V^2, Vth = 0.09 V, lambda = 0.0067
gate stack 50 nm Al2O3, Cox = 156 nF/cm^2, self-aligned S/D (0.5 um)
supply VDD = 5.0 V
return isolated per channel
tanks 2 x 50 nH per channel, 0.5 Ohm per branch

Result per channel (AC, zoomed)

ch	target	f0 [MHz]	err	depth	BW [kHz]	Q_L	ok
1	50.00	50.0002	0.00%	23.81	15.74	3176	OK
2	66.67	66.6667	-0.00%	11.97	29.94	2227	OK
3	83.33	83.3332	-0.00%	7.76	46.19	1804	--
4	100.00	99.9954	-0.00%	6.14	61.23	1633	--
5	116.67	116.6498	-0.01%	4.92	72.70	1604	--
6	133.33	133.2741	-0.04%	4.08	76.97	1732	--
7	150.00	150.0198	0.01%	3.14	40.29	3723	--
8	166.67	166.7136	0.03%	2.83	nan	nan	--
9	183.33	182.5311	-0.44%	3.94	110.22	1656	--
10	200.00	198.4061	-0.80%	3.77	114.25	1737	--

2 of 10 channels meet depth > 3 dB and BW <= 30 kHz

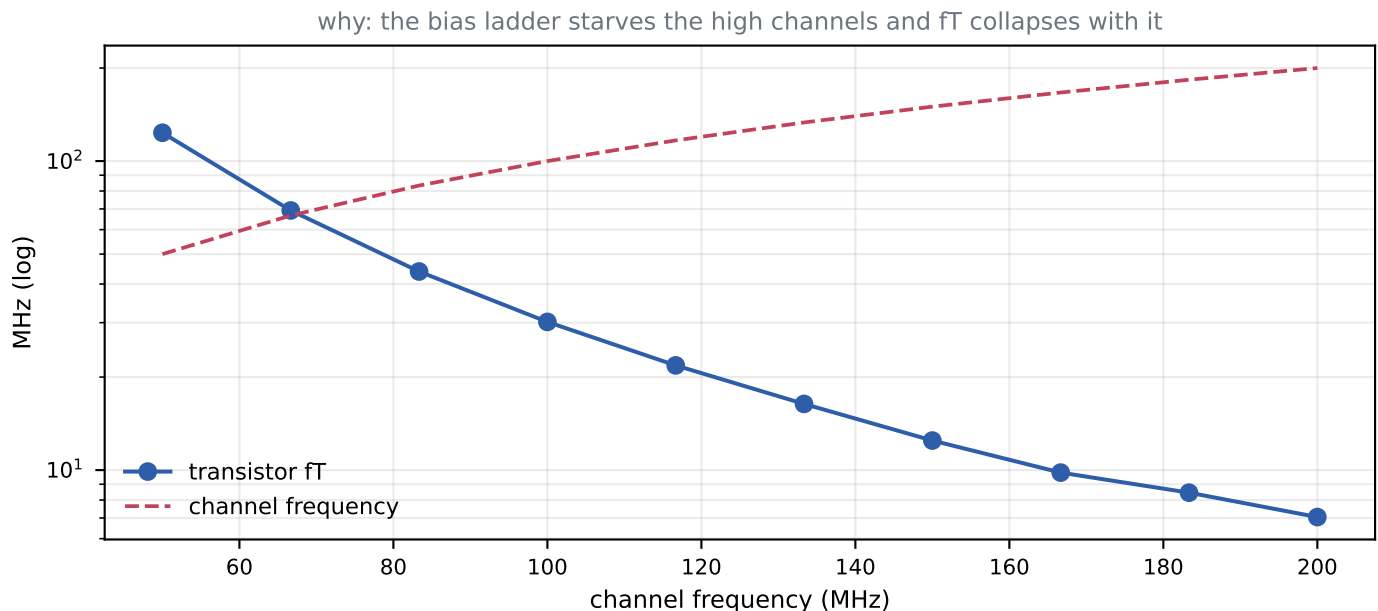
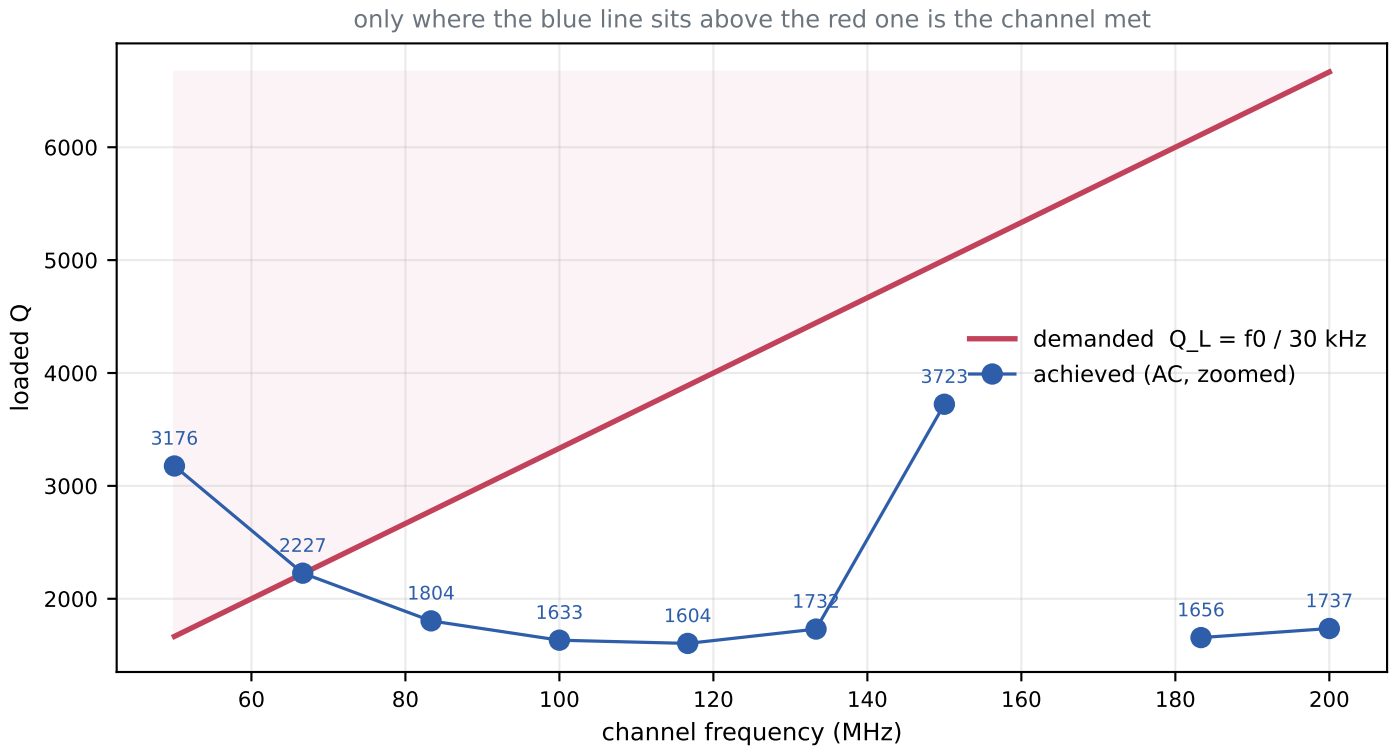
Time-domain confirmation

record 25 MB text, uniform 500 ps sampling
noise source white, new value every 0.5 ns (flat past 200 MHz)
resonances recovered from the time record: 50
spectrum computed from transient_k1000.txt

Every number here is measured, not calculated. The AC figures come from zoomed sweeps (a full-span sweep under-reports depth). The time-domain figures come from a transient driven by white noise, analysed without using .ac at all.

Quality factor: achieved against demanded

the specification asks for Q growing with frequency; the circuit delivers Q flat



The specification demands the same enhancement at every channel: $Q_L = f_0/30 \text{ kHz}$ and the bare tank $Q_0 = wL/RS$ both grow linearly with f , so each channel needs its loss cancelled to the same $\sim 53x$. That needs the transistor to still have gain at its own channel frequency.

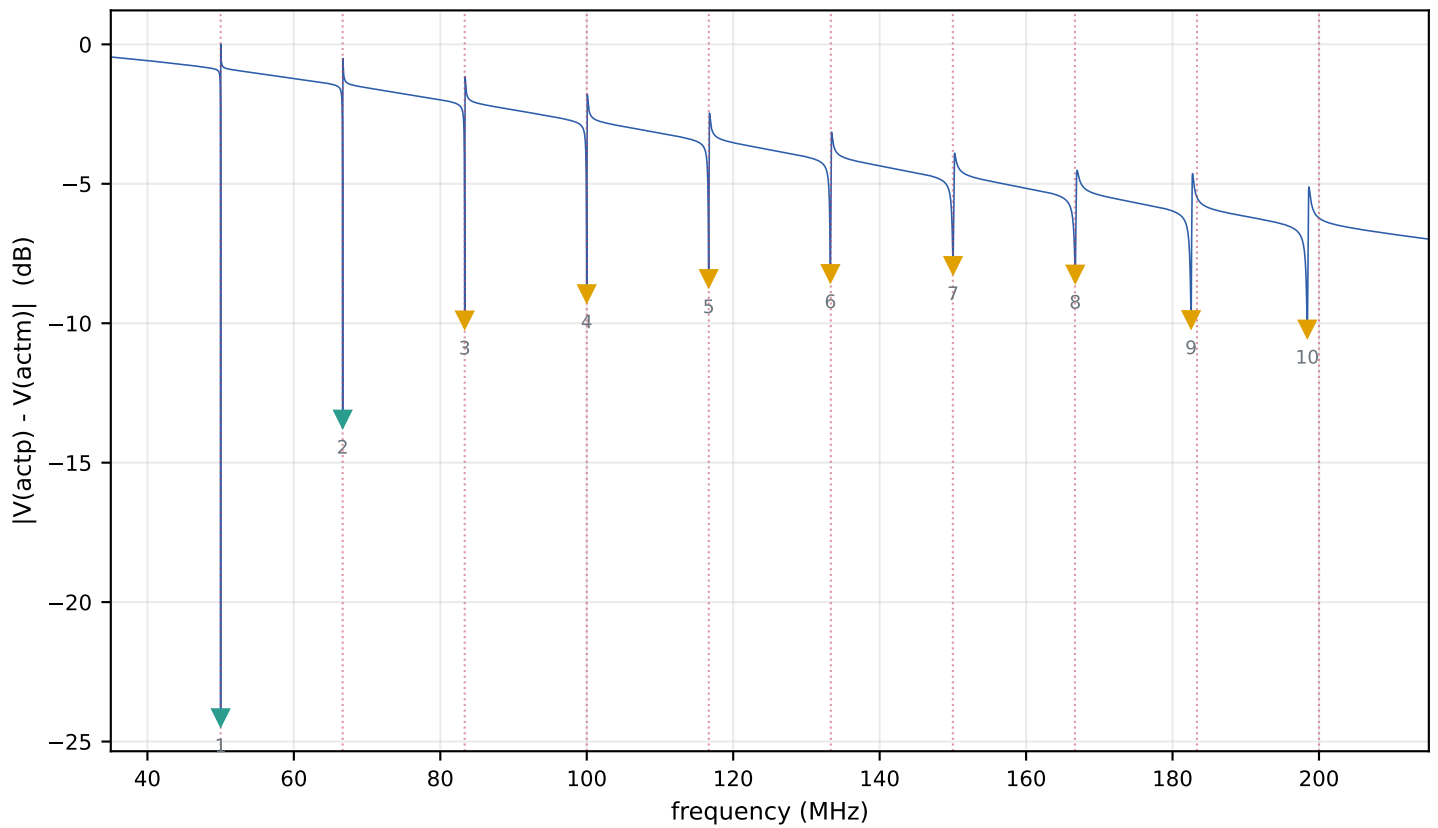
With one transistor size shared by all ten it cannot. The tank needs LESS gm as frequency rises ($gm_{req} = 2*RS/(w*L)^2$: 2026 μS down to 127 μS), so the bias ladder starves the high channels - overdrive falls 2.82 V to 0.16 V and f_T falls with it, 123 MHz to 7 MHz. Channel 10 asks a transistor 28x slower than itself to cancel its loss.

Sizing each channel for a COMMON overdrive removes the cause: $W/L = gm_{req}(f)/(K_p*V_{ov})$ makes W fall as $1/f^2$ and holds f_T at 123 MHz for all ten. Simulated: placement within 0.02 % across 50-200 MHz on the first pass, and Channel 9 rises from 1000 uS to 123 uS and f_T is 5869 against a 6111 specification.

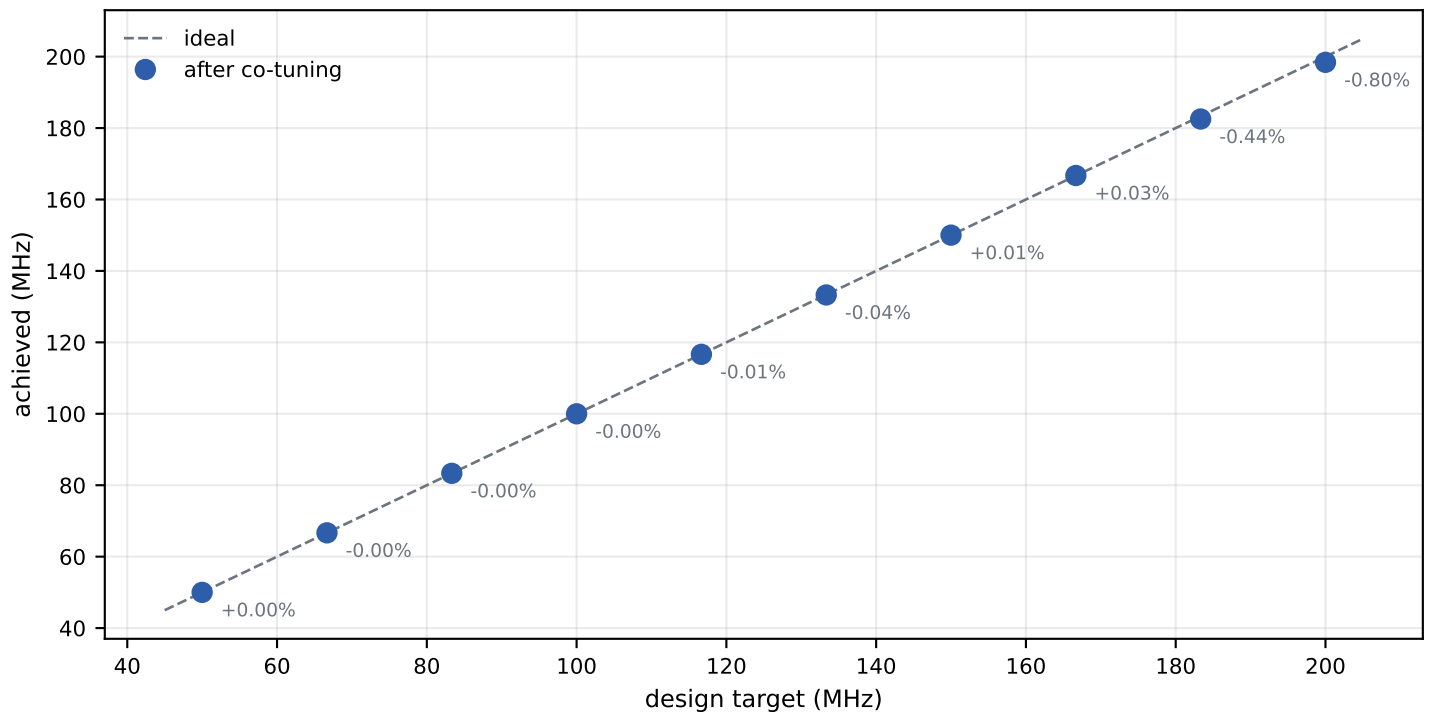
Full span, 35 - 215 MHz

all ten channels live at once; depth here is under-reported by the step size

dotted red: design targets markers: notches found

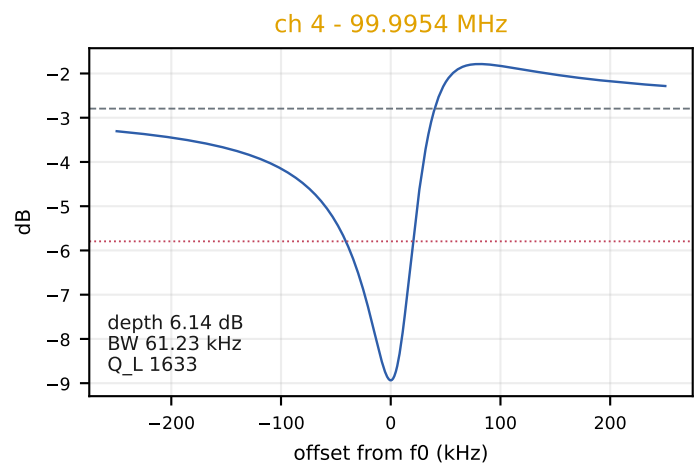
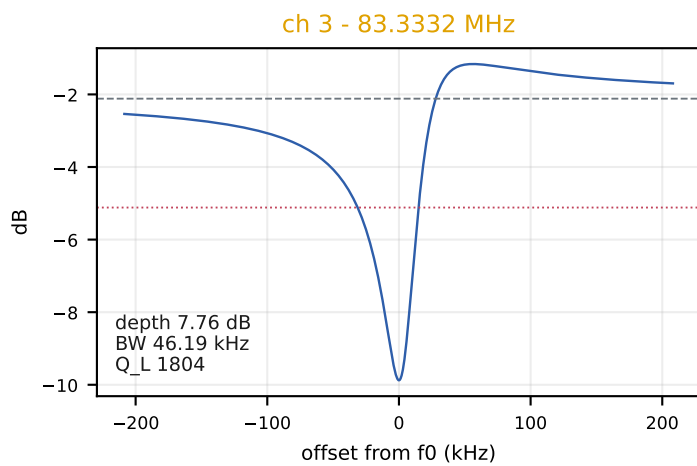
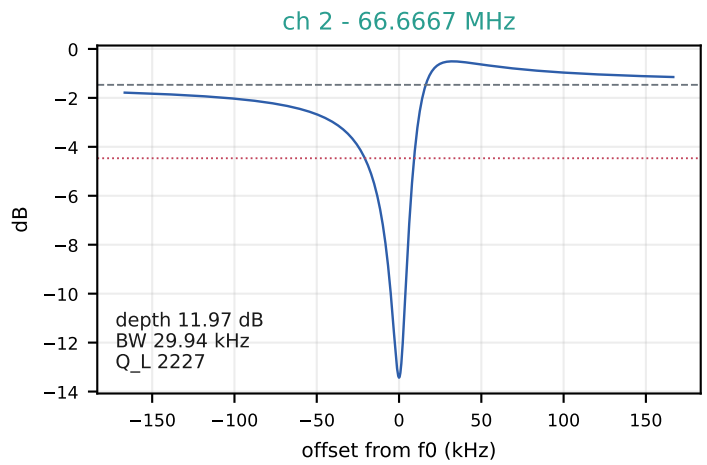
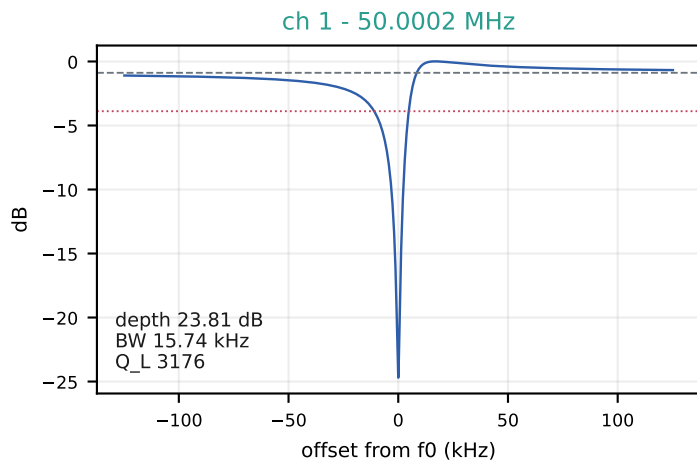


channel placement after per-channel tank trim



Per-channel zooms (1 - 4 of 10)

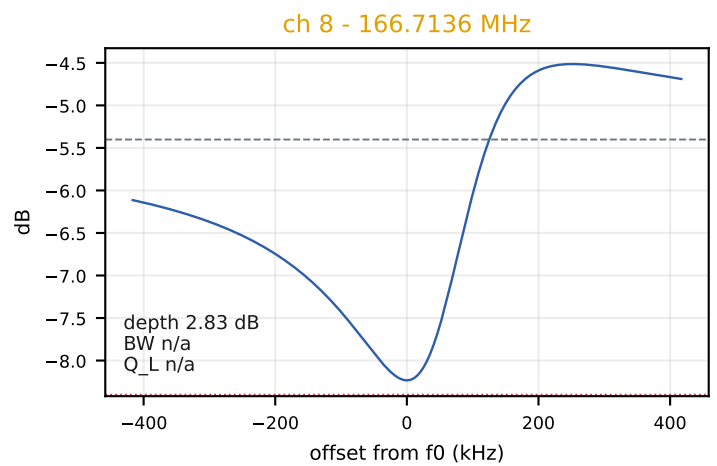
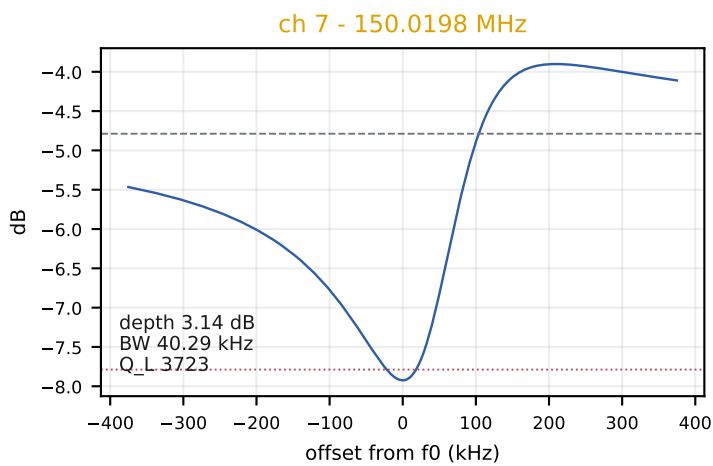
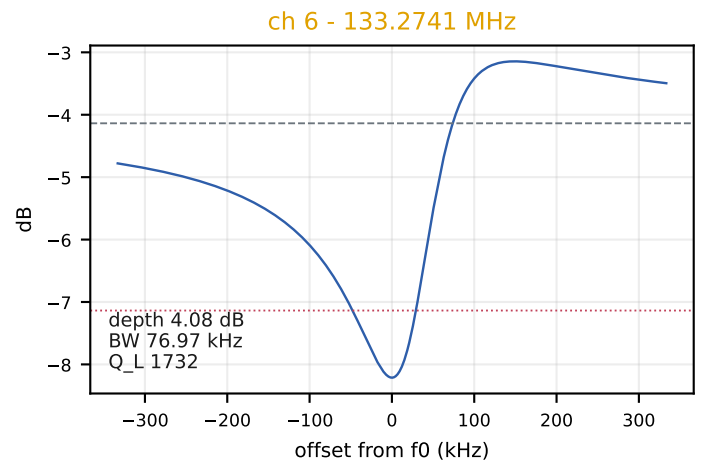
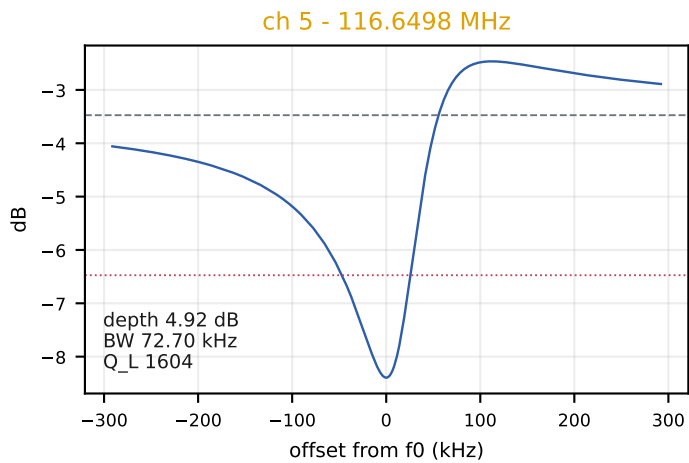
depth and -3 dB bandwidth read off the zoomed sweep, which is the only valid way to quote Q



Dashed grey: local baseline. Dotted red: baseline - 3 dB, the level the -3 dB bandwidth is measured at. A channel counts as met only when the notch is deeper than 3 dB (below that a Q number is an artefact) and the bandwidth is at or under 30 kHz.

Per-channel zooms (5 - 8 of 10)

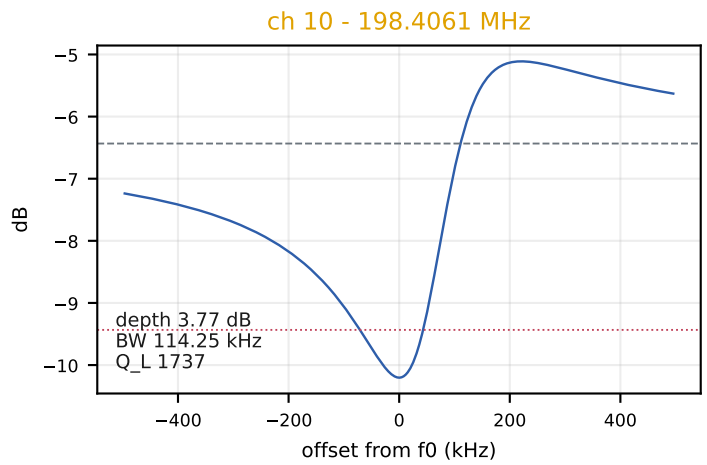
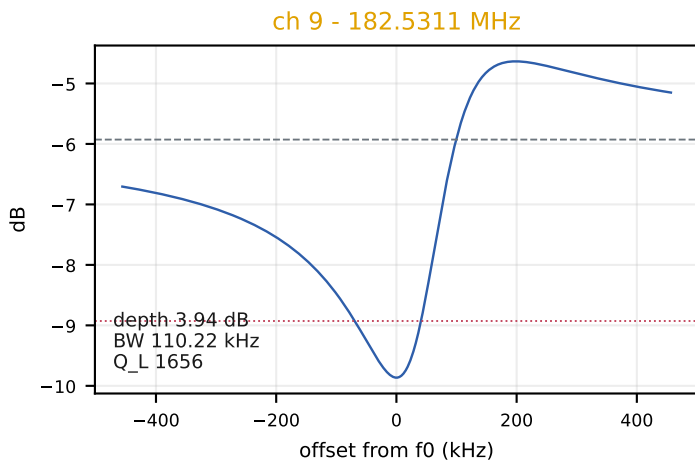
depth and -3 dB bandwidth read off the zoomed sweep, which is the only valid way to quote Q



Dashed grey: local baseline. Dotted red: baseline - 3 dB, the level the -3 dB bandwidth is measured at. A channel counts as met only when the notch is deeper than 3 dB (below that a Q number is an artefact) and the bandwidth is at or under 30 kHz.

Per-channel zooms (9 - 10 of 10)

depth and -3 dB bandwidth read off the zoomed sweep, which is the only valid way to quote Q

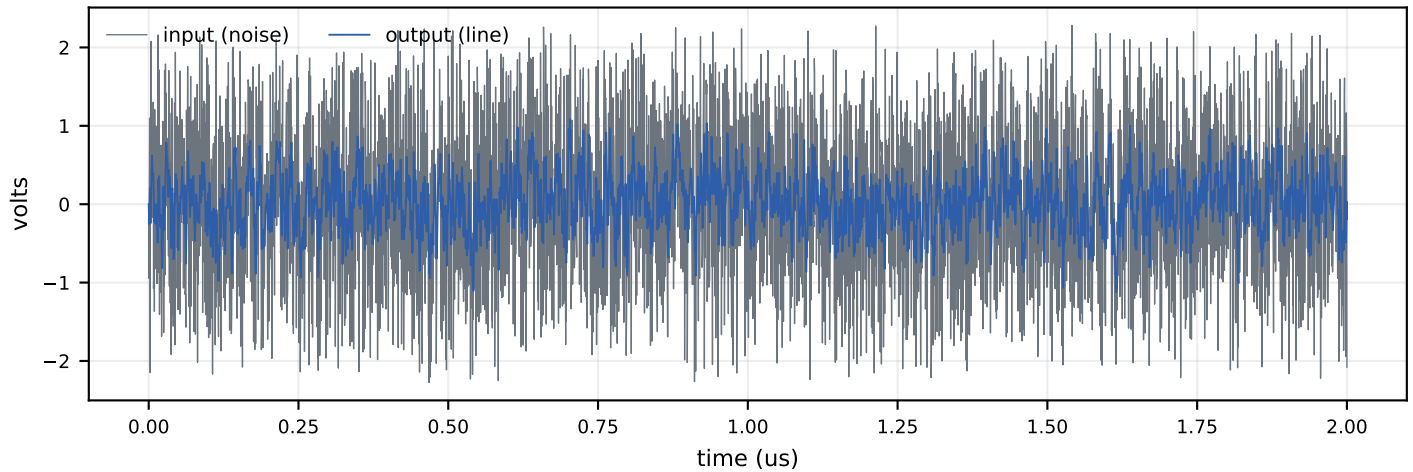


Dashed grey: local baseline. Dotted red: baseline - 3 dB, the level the -3 dB bandwidth is measured at. A channel counts as met only when the notch is deeper than 3 dB (below that a Q number is an artefact) and the bandwidth is at or under 30 kHz.

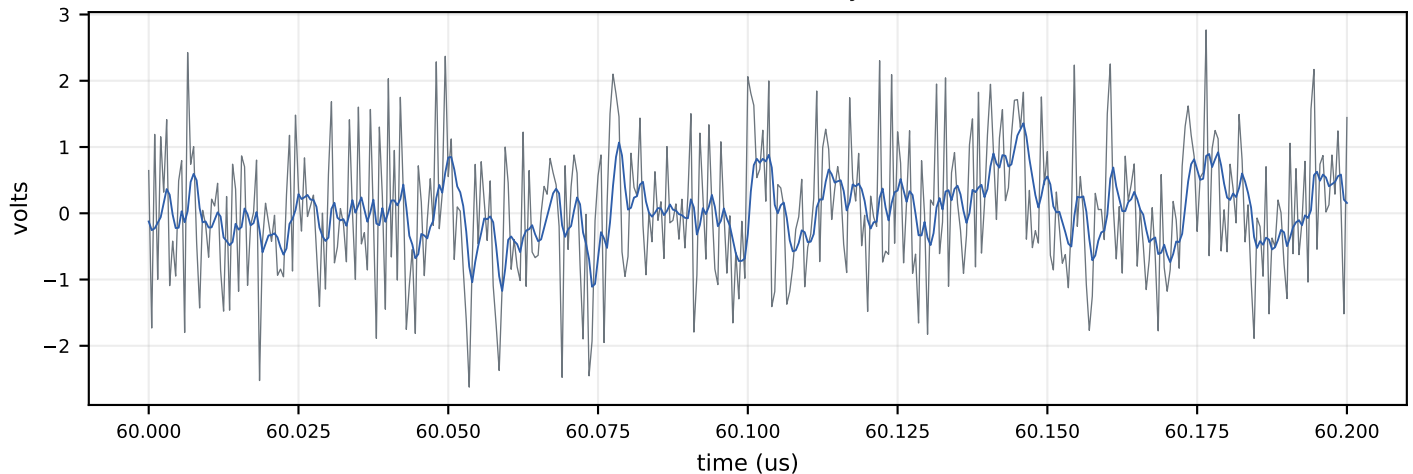
Time domain: white-noise excitation

the same circuit driven by broadband noise, no .ac involved

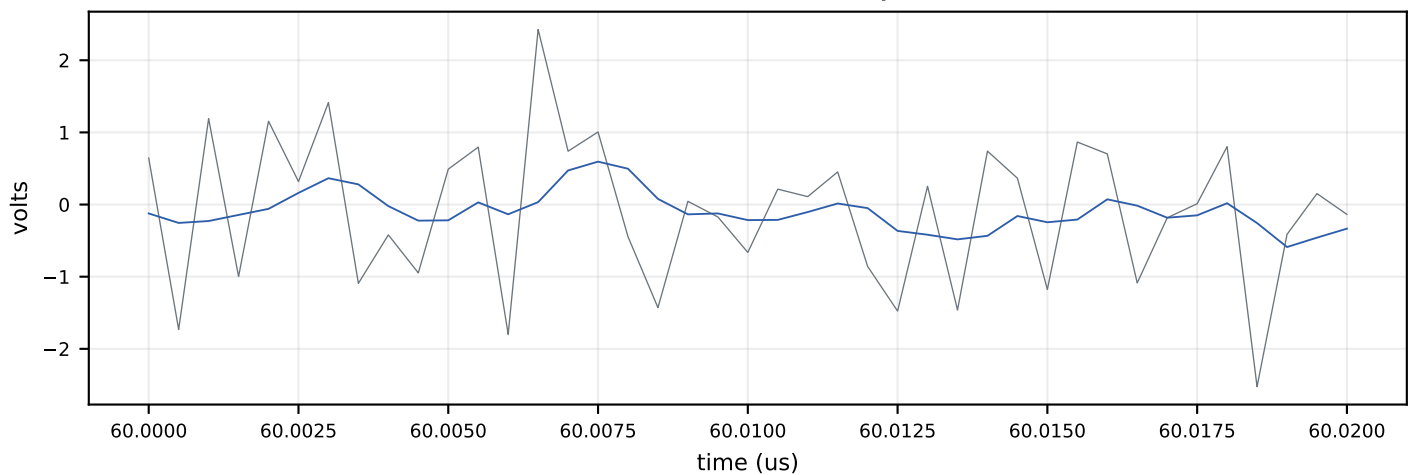
first 2 us - the tanks are still filling



200 ns window in steady state



20 ns window - individual cycles

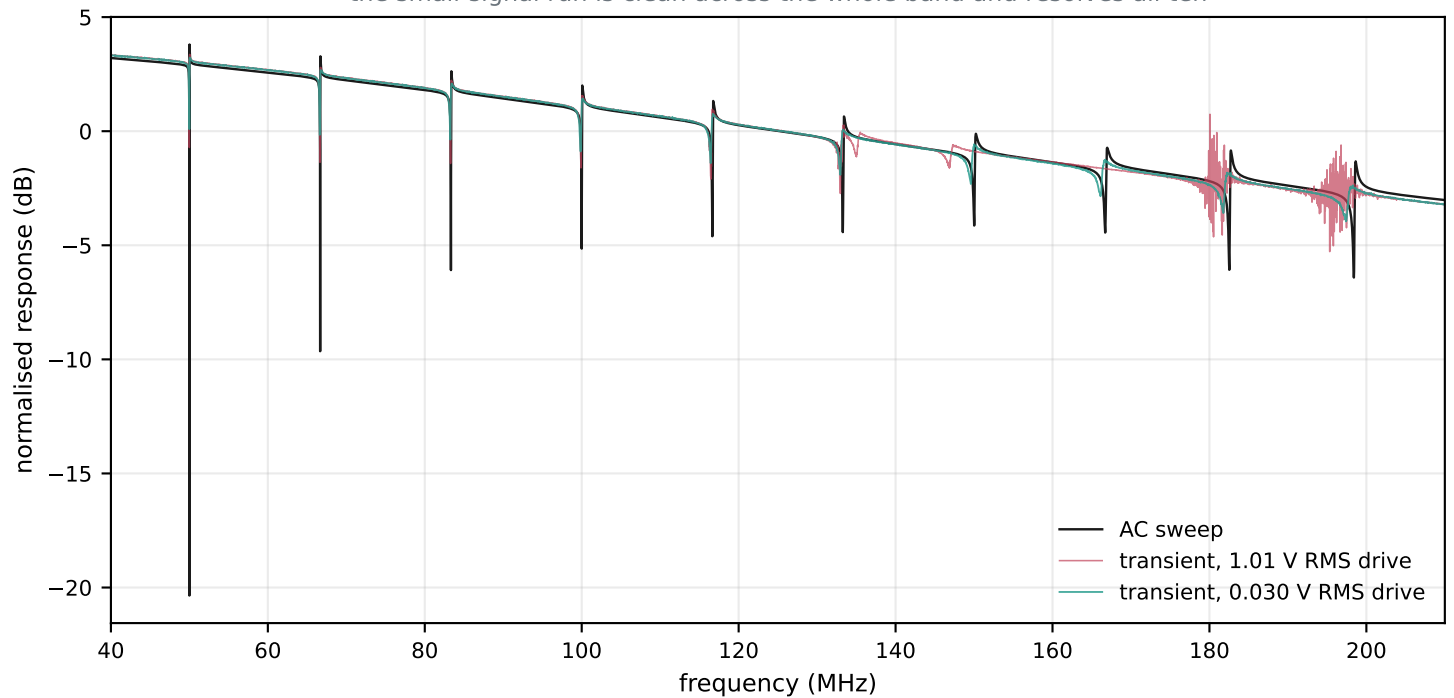


plotted straight from transient k1000.txt
input RMS 1.006 V, output RMS 0.3796 V.
The output is the input shaped by the ten resonators. Nothing about the notches is visible by eye in the time record - that is the point of the spectral estimate on the next page.

AC against the time domain, at two drive levels

same circuit, three independent measurements

the small-signal run is clean across the whole band and resolves all ten



ch	AC f0 [MHz]	1.01 V	dif kHz	0.030 V	dif kHz	AC depth
1	50.0002	49.9808	-19.4	49.9732	-27.0	23.81
2	66.6667	66.6332	-33.5	66.6131	-53.6	11.97
3	83.3332	83.2760	-57.2	83.2464	-86.8	7.76
4	99.9954	99.8855	-109.9	99.8330	-162.4	6.14
5	116.6498	116.4569	-192.9	116.4263	-223.5	4.92
6	133.2741	132.9616	-312.5	132.9462	-327.9	4.08
7	150.0198	150.5758	556.0	149.5928	-427.0	3.14
8	166.7136	167.3091	595.5	166.1194	-594.2	2.83
9	182.5311	182.2281	-302.9	181.9327	-598.3	3.94
10	198.4061	198.2376	-168.4	197.8193	-586.7	3.77

Ring-down cross-check

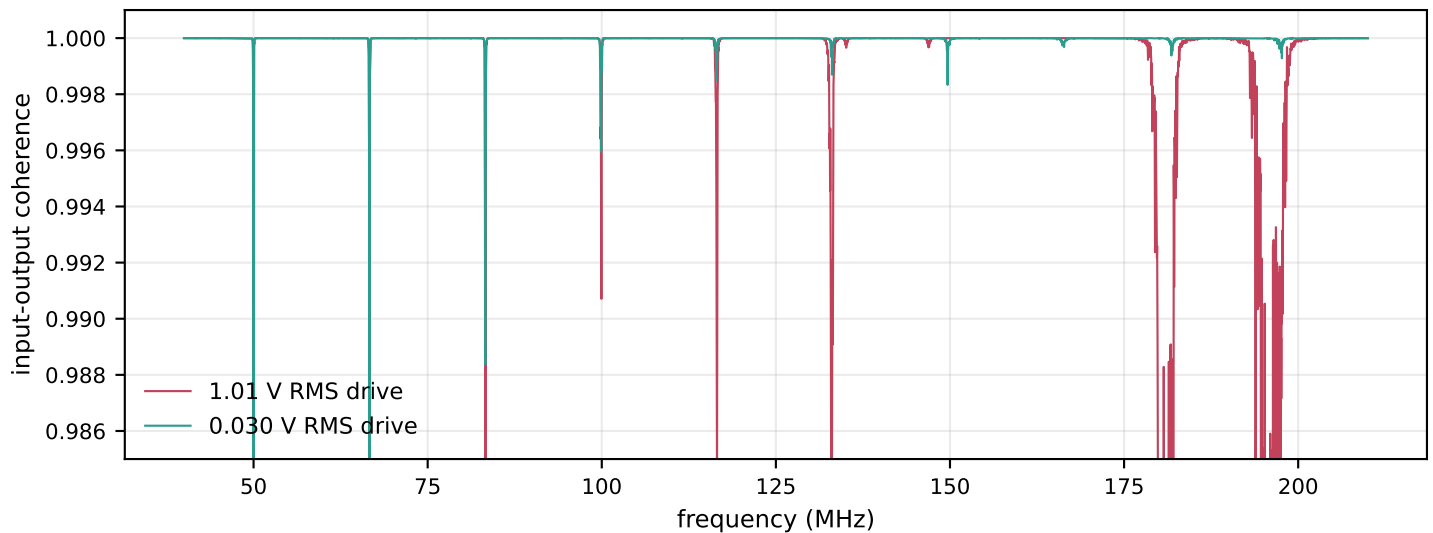
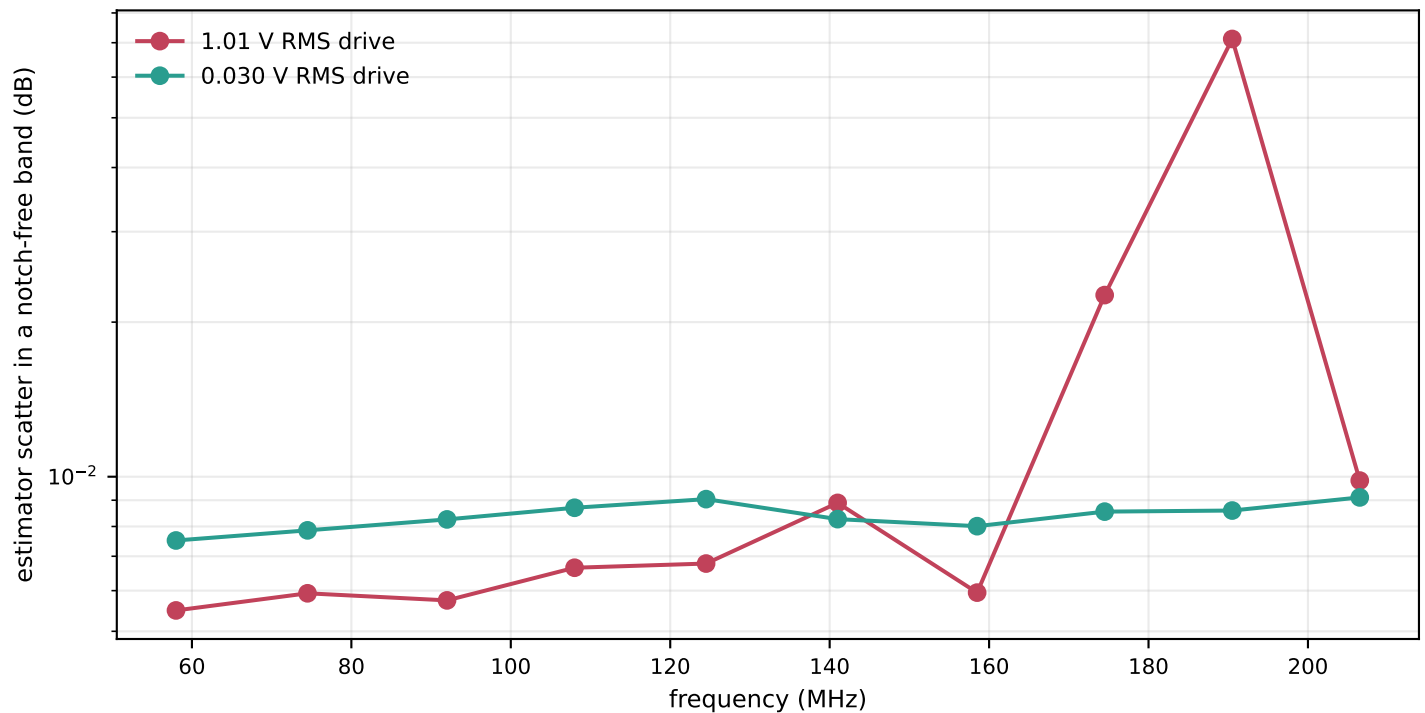
A 2 ns current pulse into the line, then nothing. The stored energy decays monotonically - RMS 2.3e-2 V over 1-3 us, 4.0e-5 over 10-25 us, 9.4e-6 over 25-50 us - so the co-tuned design point is stable, not sitting past the oscillation boundary. The ringing is dominated by 49.91 and 66.44 MHz, the first two channels. From the decay, $\tau = 13.7 \text{ us}$ gives $Q = \pi f_0 \tau = 2150$ at 49.9 MHz, against 3176 from the AC zoom: the same order, 32 % apart.

UNRESOLVED. A single-tone probe with synchronous detection, swept across channel 1 and channel 5, shows only 0.10 and 0.25 dB of variation where the AC predicts 24 and 4.9 dB. Drive amplitude, timestep, settling time and estimator noise were each tested and ruled out. The noise-driven spectrum and the ring-down both support the AC, the tone probe does not, and the reason is not yet identified. Treat the absolute depths as provisional until it is.

Why the transient spectrum frays above channel 6

measured, by re-running the same circuit at 1/34 the drive amplitude

the excess is only there at large drive, and only where channels 8 and 9 sit



The cross-coupled pairs are not linear, and at 1 V RMS they intermodulate. The channel plan makes that land badly: the ladder is arithmetic, $50 \text{ MHz} + k \times 16.67$, so the sum of any two channels is another channel. Channel 9 at 183.3 MHz gets $\text{ch1}+\text{ch6}$, $\text{ch2}+\text{ch5}$ and $\text{ch3}+\text{ch4}$; channel 8 at 166.7 MHz gets $\text{ch1}+\text{ch5}$, $\text{ch2}+\text{ch4}$ and twice ch3 . That energy is not linearly related to the input, so coherence drops and the estimate frays - exactly above channel 6, where it was observed.

Re-running at 0.030 V RMS removes it: scatter falls from 0.071 dB to 0.009 dB at 187-194 MHz and from 0.023 dB to 0.008 dB at 171-178 MHz, and coherence returns to 1.0000 across the band. Above channel 10, at 203-210 MHz, the large-signal run is already clean - there is no channel there to receive a sum.

A measurement artefact here, but a real design warning: an evenly spaced channel plan puts every second-order product on top of another channel. At 1000 channels that is a crosstalk mechanism; non-arithmetic spacing would avoid it.

Ten-channel ladder, $W/L = 1000$ - AC and transient